Chapter II
Can a machine think?

Overview

The inquiry "Can a machine think?" is probably the most essential and persistent philosophical question in the domain of Artificial Intelligence. It was initially and meticulously examined by computer pioneer Alan Turing in his key 1950 article, "Computing Machinery and Intelligence."

Objectives

•To investigate the notion of cognition and its implications in both human and artificial realms.

•To analyse Alan Turing's inquiry and the Turing Test,

•To comprehend the distinctions between human intelligence and artificial intelligence.

•To examine philosophical arguments on the feasibility of machine cognition,

•To examine fundamental viewpoints from philosophers and scientists, including Alan Turing, John Searle

In the early 1950s, the ambiguous inquiry arose: might a machine possess the capacity for thought? was substituted with the more accessible inquiry: might a computer that manipulates physical symbols in accordance with structure-sensitive rules contain the capacity for thought? This inquiry represents a progression resulting from extensive study in formal logic and computational theory during the past fifty years. Theorists recognized the significant influence of abstract symbolic systems that withstand rule-based modifications. If those systems were automated, their abstract computational skills would manifest in a tangible physical system. This discovery initiated a well-defined research initiative with significant theoretical foundations. Three individuals are distinguished for establishing the intellectual and technical foundations of the discipline known as Artificial Intelligence: John von Neumann, Alan Turing, and Claude Shannon.

Yes, it can

Could a machine think? There were many reasons for saying yes. One of the fundamental and profound causes is in two significant findings in computational theory. The first was Church's thesis, states that every effectively computable function is recursively computable. Effectively computable means that there is a "rote" procedure for determining, in finite time, the output of the function for a given input. Recursively computable means more specifically that there is a finite set of operations that can be applied to a given input, and then applied again and again to the successive results of such applications, to yield the function's output in finite time. The notion of a rote procedure is nonformal and intuitive; thus, Church's thesis does not admit of a formal proof. But it does go to the heart of what it is to compute, and many lines of evidence converge in supporting it.

Alonzo Churc posited that a function is effectively computable if and only if it can be calculated by lambda calculus, a formal language he devised to express mathematical functions and computation. Simultaneously, Turing conceptualised the Turing machine, an abstract device that processes symbols on a tape in accordance with predefined rules.

Church's Thesis is not a formal theorem subject to proof; instead, it constitutes a hypothesis or philosophical assertion regarding the essence of computation. It links the intuitive concept of an algorithm with exact mathematical definitions. The thesis constitutes the cornerstone of computer science, as all contemporary computers are fundamentally physical embodiments of Turing machines.

The thesis presents significant philosophical enquiries concerning the nature of the mind and artificial intelligence. If human cognition adheres to computable processes, then, theoretically, a machine could emulate human reasoning. Nonetheless, if the mind executes non-computable operations—transcending algorithms—machines may never entirely replicate human consciousness or creativity.

The second important result was Alan M. Turing's demonstration that any recursively computable function can be computed in finite time by a maximally simple sort of symbol-manipulating machine that has come to be called a universal Turing machine. This machine is

guided by a set of recursively applicable rules that are sensitive to the identity, order and arrangement of the elementary symbols it encounters as input.

Alan Turing In a significant study titled “Intelligent Machinery,” Turing examined the requirements for replicating intelligence within a machine, a formidable challenge considering the technological constraints of that era. “The human brain's memory capacity is likely around ten billion binary digits,” he contemplated. “However, a significant portion of this is likely utilized for recalling visual impressions and other relatively inefficient methods.” One could justifiably anticipate achieving significant advancements in Artificial Intelligence with several million digits of computer memory.
He proposed the Turing Test – a method for assessing a machine's intelligence by determining if its responses can be differentiated from those of a human during conversation. If a human interrogator could not consistently distinguish between the two, the computer may be considered to "think." The Turing Test emerged as a pivotal concept in the philosophy of artificial intelligence, inciting much discourse regarding consciousness, cognition, and the essence of thought

The inaugural commercial scientific computer and mathematician Claude Shannon, the progenitor of information theory. The purpose of DSRPAI (Dartmouth Summer Research Project on Artificial Intelligence) was to gather academics from diverse disciplines to establish a novel research domain focused on developing machines capable of simulating human intelligence.

John von Neumann’s major contribution to computing was helping to establish the idea of a computer program store in the computer memory. Von Neumann was, in fact, the first person to use the human terminology “memory” when referring to a computer. Unlike some of his contemporaries, he did not believe a computer would be able to think in the way that a human can, but he did help establish the parallels that exist with human physiognomy. The parts of a computer, he wrote in one paper, “correspond to the associative neurons in the human nervous system. It remains to discuss the equivalents of the sensory or afferent and the motor or efferent neurons.” Others would happily take up the challenge.

These two results entail something remarkable, namely that a standard digital computer, given only the right program, a large enough memory and sufficient time, can compute any rule-

governed input-output function. That is, it can display any systematic pattern of responses to the environment whatsoever.

More specifically, these results imply that a suitably programmed symbol-manipulating machine (SM machine) should be able to pass the Turing test for conscious intelligence. The machine passes this test for conscious intelligence if its responses cannot be discriminated from the typewritten responses of a real, intelligent person. the Church and Turing results assure us that, whatever that (presumably effective) function might be, a suitable SM machine could compute it.

The preliminary outcomes were favourable. symbol-manipulating machine robots equipped with sophisticated algorithms executed a range of seemingly intelligent tasks. They addressed intricate instructions, resolved advanced arithmetic, algebraic, and strategic issues, participated in checkers and chess, demonstrated theorems, and engaged in basic discussion. Performance consistently enhanced with the emergence of higher memory capacities, accelerated machines, and the implementation of more extensive and sophisticated programs. Classical, or "program-writing," AI represented a robust and fruitful research endeavor from nearly every standpoint. The sporadic assertion that a superintelligent machine may never get cognitive capabilities seemed ignorant and poorly motivated. The argument for an affirmative response to our titular inquiry seemed compelling.

It cannot

Could a machine think? There were also some arguments for saying no. The objection was occasionally made that thinking was a nonphysical process in an immaterial soul. But such dualistic resistance was neither evolutionarily nor explanatorily plausible. It had a negligible impact on Ai research. A quite different line of objection was more successful in gaining the Ai community's attention. In 1972 Hubert L. Dreyfus published a book that was highly critical of the parade-case simulations of cognitive activity. He contended their insufficiency as representations of authentic cognition and highlighted a recurring pattern of failure in these endeavors. He proposed that they lacked the extensive reservoir of unexpressed prior information inherent in every individual, as well as the intuitive ability to utilize pertinent elements of that knowledge in response to evolving situations. Dreyfus did not deny the possibility that an artificial physical system of some kind might think, but he was highly critical of the idea that this could be achieved solely by symbol manipulation at the hands of recursively applicable rules.

John Searle authored a new and quite different criticism aimed at the most basic assumption of the classical research program: the idea that the appropriate manipulation of structured symbols by the recursive application of structure sensitive rules could constitute conscious intelligence. in his 1980 paper “Minds, Brains and Programs.” is designed to demonstrate that the software cannot make a computer conscious or give it a mind that is anything like a human mind. Searle's paper provoked a lively reaction from AI researchers, psychologists and philosophers alike. On the whole, however, he was met with an even more hostile reception than Dreyfus had experienced. The general lesson drawn is that any system that merely manipulates physical symbols in accordance with structure-sensitive rules will be at best a hollow mock-up of real conscious intelligence, because it is impossible to generate "real semantics" merely by cranking away on "empty syntax. "

Demonstrate what you obtained

1 Identify the philosophers who contend that machines possess the ability to think.

2 Examine Church's thesis.

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3 Compose a written piece regarding Alan Turing.

Alan M. Turing's demonstration that any recursively computable function can be computed in finite time by a maximally simple sort of symbol-manipulating machine that has come to be called a universal Turing machine. This machine is guided by a set of recursively applicable rules that are sensitive to the identity, order and arrangement of the elementary symbols it encounters as input

4 Compose a piece regarding Claude Shannon. Alan M. Turing's Alonzo Churc

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Claude Shannon. Hubert L. Dreyfus.

5 Compose a piece on Hubert L. Dreyfus.

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Knock your Brain

1--------- authored the article “Minds, Brains and Programs.”

(Claude Shannon. Alan M. Turing's John Searle)

2 Who was to use the human terminology “memory” when referring to a computer

(Claude Shannon. John von Neumann, Alonzo Church)

3 Who is the founder of information theory?

(Claude Shannon. Alan M. Turing, Alonzo Church)

4 The author of the notable research entitled “Intelligent Machinery,”

(Claude Shannon. Alan M. Turing, Alonzo Church)

Unleash your potential

1. Analyze the notion Can a machine think?

2. Critically evaluates the concept Can a machine think